Zepto Inventory & Pricing Analysis

Objective

Zepto is a quick-commerce grocery platform where margins are thin and shelf space in dark stores is limited. This project analyses a product-level dataset to answer four business questions:

- Where are the biggest discounts, and are they being used wisely?
- Which high-value products are going out of stock?
- Which categories hold the most inventory value and weight?
- Which products offer customers the best value per gram?

About the dataset

This dataset provides a detailed view of the product catalog and pricing structure of Zepto, a fast-growing 10-minute grocery delivery platform. The data captures essential attributes for over 3,000+ SKUs (Stock Keeping Units) across various categories like Fruits & Vegetables, Dairy, Packaged Foods, Beverages, and more.

The data is structured to support various types of retail analysis, including:

- Discount trends by category
- Inventory availability and stock-outs
- Price distribution and pricing strategy
- Product naming patterns

Data Exploration and Cleaning

Exploration steps

- Counted the total number of rows in the table.
- Viewed a 10-row sample to confirm the structure.
- Checked every column for null values.
- Listed the distinct product categories in sorted order.
- Compared the number of in-stock and out-of-stock products.
- Looked for product names that appear more than once under different SKUs.

Cleaning steps

- Found rows where MRP or selling price was 0 and deleted records with MRP = 0, since they carry no usable pricing information:

- Prices in the raw file were stored in paise, so both `mrp` and `discountedSellingPrice` were divided by 100 to convert to rupees.

Analysis and Findings

We performed structured analysis in MySQL to answer key business questions:

1. **Best Discounts;** Find the top 10 best-value products based on the discount percentage.

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Orange Wafers	45.00	51.00
2	Dukes Waffy Chocolate Wafers	45.00	51.00
3	Dukes Waffy Strawberry Wafers	45.00	51.00
4	RRO Mozzarella Block Cheese	295.00	50.00
5	Ceres Foods Laal Maas Instant Liquid Mas...	220.00	50.00
6	Chef's Basket Durum Wheat Elbow Pasta	160.00	50.00
7	RRO Mozzarella Pizza Cheese	275.00	50.00
8	Ceres Foods Fish Mustard Instant Liquid M...	220.00	50.00

2. Out-of-Stock Premium - What are the products with high MRP (above ₹300) that are out of stock?

	name character varying (150)	mrp numeric (8,2)
1	Patanjali Cow's Ghee	565.00
2	MamyPoko Pants Standard Diapers, Extra Large (12 - 17 ...	399.00
3	Aashirvaad Atta With Mutigrains	315.00
4	Everest Kashmiri Lal Chilli Powder	310.00

3. Category revenue - What is the estimated revenue for each category?

	category character varying (120)	total_revenue numeric
1	Fruits & Vegetables	10846.00
2	Meats, Fish & Eggs	20693.00
3	Biscuits	25007.60
4	Beverages	55051.00
5	Dairy, Bread & Batter	55051.00
6	Health & Hygiene	64180.00
7	Home & Cleaning	122661.00
8	Ice Cream & Desserts	224385.00

4. High Price, Low Discount - Which products have an MRP greater than ₹500 and a discount of less than 10%?

	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dhara Kachi Ghani Mustard Oil Jar	1250.00	8.00
2	Saffola Gold (Jar)	1240.00	0.00
3	Dhara Filtered Groundnut Oil (Jar)	1050.00	1.00
4	Fortune Rice Bran Health Oil (Jar)	1050.00	1.00
5	Dhara Filtered Groundnut Oil (Jar)	1050.00	0.00
6	Fortune Soyabean Oil	1005.00	0.00
7	Fortune Sunlite Refined Sunflower (Jar)	925.00	0.00
8	Surf Excel Matic Powder Front Load	810.00	7.00

5. Top Discount Categories - Which top 5 categories offer the highest average discount percentage?

	category character varying (120)	avg_discount numeric
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

6. Price per Gram - What is the price per gram for products of 100 g and above, sorted by best value?

	name character varying (150)	weightingms integer	discountedsellingprice numeric (8,2)	price_per_gram numeric
1	Tata Salt	1000	24.00	0.02
2	Onion	1000	21.00	0.02
3	Onion	3000	57.00	0.02
4	Shubh kart - Nirmal sugandhi mogra wet dhoop zipper 20 sticks	1160	28.00	0.02
5	Vicks Cough Drops Menthol	1160	20.00	0.02
6	Aashirvaad Iodised Salt	1000	19.00	0.02
7	Carrot	500	15.00	0.03
8	Potato	1000	29.00	0.03

7. Weight Segments - How can the products be grouped into Low, Medium and Bulk categories?

	name character varying (150)	weightingms integer	weight_category text
1	Popular Essentials Poppy Seeds (Gasagase)	100	Low
2	Origami Kitchen Towels 2 In 1 Quilted 2ply	116	Low
3	Origami Good Karma Paper Serviettes - 100 napkins (1 ply)	58	Low
4	Johnson's Baby Top To Toe Bath Body Wash	210	Low
5	Harpic Hygienic Toilet Cleaner Rim Block Citrus	26	Low
6	Lizol Disinfectant Surface & Floor Cleaner Liquid Jasmine	500	Low
7	Sapota	348	Low
8	Britannia Treart Creme Wafers Chocolate	60	Low

8. **Inventory weight** - What is the total inventory weight per category?

	category character varying (120)	total_weight bigint
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187
8	Personal Care	348187

Business Recommendations

1. Prioritise restocking of high-MRP out-of-stock products . Premium items that run out lose the most revenue per missed order. Set higher safety-stock levels and faster replenishment triggers for SKUs above a price threshold such as ₹300, and track their stock-out rate as a separate KPI.

2. Audit the deepest discounts . Check whether the highest-discount products and categories actually lift volume. Where a deep discount does not increase sales, reduce it and recover margin. Where it is clearing near-expiry stock, make sure that is the stated purpose and the discount ends once the stock is gone.

3. Use targeted, not blanket, discounting. Categories with high average discounts should be tested against categories with low discounts and similar sales. Shift promotion budget toward categories where discounts demonstrably drive repeat orders.

4. Protect margin on premium, low-discount products . Products priced above ₹500 with under 10% discount are candidates for premium positioning. Keep them well stocked and visible. If any are slow movers, trial a modest discount before they become dead stock.

5. Focus inventory planning on the highest-value categories. Categories that hold the most potential revenue need the tightest demand forecasting. Over-stocking ties up capital, and under-stocking loses sales.

6. Plan logistics around heavy categories . Categories with high total weight need more storage and costlier delivery. Consider minimum order values, delivery fees or bundling rules for bulk and heavy items so they do not erode margin on small orders.

7. Use price-per-gram to guide assortment and pack sizes . Customers compare unit prices. Promote the best-value pack sizes, and review products with very high price per gram to confirm they justify their premium (brand, quality, organic, etc.).

8. Clean up duplicate and inconsistent listings. Names appearing under multiple SKUs should be reviewed for genuine variants vs listing errors, which can confuse customers and distort inventory counts.